



Okan Berhoğlu

Robotics Engineer

+90 535 063 70 01 | okanberh@gmail.com

<https://okanberhoglu.github.io/>

PROFILE

Robotics Engineer experienced in desktop and industrial software development, machine data acquisition, device communication, and engineering systems. Skilled in communication protocols, database integration, deployment, validation, and technical documentation. Background in Mechanical Engineering and ongoing M.S. studies in Robotics.

EDUCATION

Middle East Technical University *Ankara, Turkey*
Master of Science, Robotics
2024 – 2026

TOBB Economics and Technology University
Ankara, Turkey
Bachelor of Science, Mechanical Engineering
2018 – 2023

Atatürk University *Remote*
Associate Degree, Computer Programming, Second University Program
2021 – 2023

SKILLS

Software: Java, C#, Python, C++, REST API, Git, Linux, CUDA, Unity, ROS/ROS2, MATLAB, Simulink, Simscape, SQL, InfluxDB

Industrial Communication: TCP/IP, UDP, EtherNet/IP, Modbus TCP, CAN, RS-485, industrial machine communication

Language: English - Upper-Intermediate, TOEFL IBT 2026 score: 5/6 (equivalent to 100/120 in old score system)

PROJECTS

Development of a Gimbal and Its Digital Twin
Designed and developed a gimbal and its digital twin enabling remote control of a physical gimbal's direction, featuring a React.js web application with a 3D model (built in Three.js) that mirrors the gimbal's movements in real-time.

Customizable Laptop Stand MATLAB App
Developed a user-friendly MATLAB app that generates a customizable 3D laptop stand model in STL format based on user-defined dimensions, using a 3D topology optimization script for the mechanical design and parametrically modeling the resulting shape with Bezier surfaces.

ChatMoonVLM

Developed a locally-run visual question answering web app that uses a vision-language model to let users upload an image and hold a conversation.

EXPERIENCE

Middle East Technical University *Ankara, Turkey*

Researcher *October 2025 - Present*

- Conduct research on mobile robotic systems, including kinematic and dynamic modeling, simulation, sensor integration, and autonomous control algorithm development.
- Contribute to a research project investigating human rhythmic behavior. Within the project, I developed an experimental setup using a 3-axis haptic device, collected data from predefined rhythmic motions, and conducted system identification and behavior modeling studies.

Teknopar Industrial Automation *Ankara, Turkey*

Software Engineer *February 2024 – October 2025, 1 year 8 months*

- Designed the software architecture and developed a Windows service-based data acquisition application for CNC controllers. Integrated TCP/UDP-based industrial communication protocols to collect real-time machine data, standardized the collected data into an MTConnect-like format, and actively contributed to the software's licensing, deployment, and commissioning processes.
- Developed an OSGi-based application with a Swing user interface for military truck systems, enabling operators to monitor onboard power units and control the vehicle's electrical system. Integrated industrial communication protocols such as EtherNet/IP and Modbus TCP for device communication and data collection.
- Actively contributed to the proposal writing of a TÜBİTAK-funded project focused on developing the system model, autonomy stack, and digital twin of a mobile robot. Within the project, worked on AI-based environmental perception and autonomous navigation using ROS2 as a software framework, 3D LiDAR, and depth camera sensors.

Candidate Engineer *December 2022 – February 2024, 1 year 3 months*

- Supported R&D activities focused on collecting data from CNC machines and industrial systems.
- Contributed to the modeling of an electromechanical system using Simulink and Simscape, and developed its integration into IBM Rhapsody.
- Analyzed experimental data from hardware prototypes and supported grey-box system identification and parameter estimation studies.

Intern *May 2022 – December 2022, 8 months*

- Worked on data acquisition methods from CNC machines.
- Built a MATLAB Simscape Multibody model to simulate an electromechanical system's mechanical behavior.
- Contributed to mechanical design by creating hydraulic system components in SolidWorks.

Anova R&D Technologies *Ankara, Turkey*

Intern *May 2021 – August 2021, 4 months*

- Gained hands-on experience in machining, quality control, and finishing processes, and contributed to mold and fixture design activities.